

Artificial Neural Network (1)

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- 1 Big Idea
- 2 Fundamentals

Big Idea

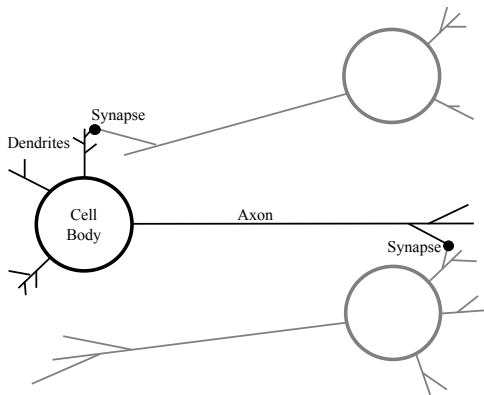


Figure 1: A high-level schematic of the structure of a neuron. This figure illustrates three interconnected neurons; the middle neuron is highlighted in black, and the major structural components of this neuron are labeled cell body, dendrites, and axon. Also marked are the synapses connecting the axon of one neuron and the dendrite of another, which allow signals to pass between the neurons.

Fundamentals

$$z = \underbrace{\mathbf{w}[0] \times \mathbf{d}[0] + \mathbf{w}[1] \times \mathbf{d}[1] + \cdots + \mathbf{w}[m] \times \mathbf{d}[m]}_{\text{weighted sum}} \quad (1)$$

$$= \sum_{j=0}^m \mathbf{w}[j] \times \mathbf{d}[j]$$

$$= \underbrace{\mathbf{w} \cdot \mathbf{d}}_{\text{dot product}} = \underbrace{\mathbf{w}^T \mathbf{d}}_{\text{matrix product}} = [w_0, w_1, \dots, w_m] \begin{bmatrix} d_0 \\ d_1 \\ \vdots \\ d_m \end{bmatrix} \quad (2)$$

$$\mathbb{M}_{\mathbf{w}}(\mathbf{d}) = \begin{cases} 1 & \text{if } z \geq \theta \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

$$\text{rectifier}(z) = \max(0, z) \quad (4)$$

Artificial Neurons

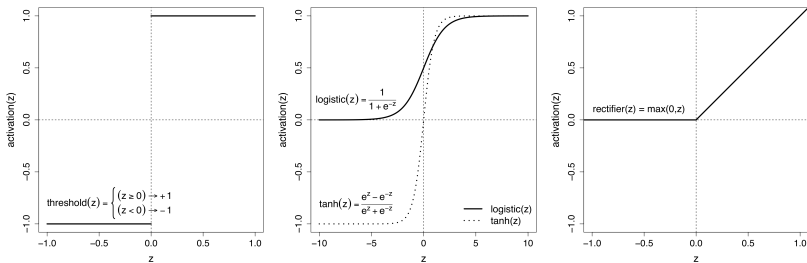


Figure 2: Plots for activation functions that have been popular in the history of neural networks.

$$\begin{aligned}\mathbb{M}_{\mathbf{w}}(\mathbf{d}) &= \varphi(\mathbf{w}[0] \times \mathbf{d}[0] + \mathbf{w}[1] \times \mathbf{d}[1] + \dots + \mathbf{w}[m] \times \mathbf{d}[m]) \\ &= \varphi\left(\sum_{i=0}^m w_i \times d_i\right) = \varphi\left(\underbrace{\mathbf{w} \cdot \mathbf{d}}_{\text{dot product}}\right) \\ &= \varphi\left(\underbrace{\mathbf{w}^T \mathbf{d}}_{\text{matrix product}}\right) = \varphi\left([w_0, w_1, \dots, w_m] \begin{bmatrix} d_0 \\ d_1 \\ d_2 \\ \vdots \\ d_m \end{bmatrix}\right) \quad (5)\end{aligned}$$

Artificial Neurons

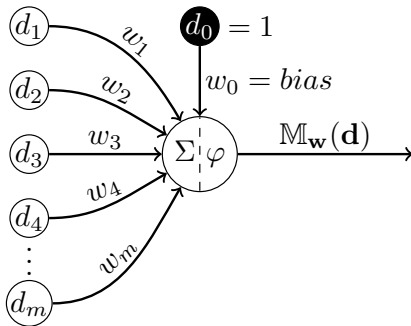


Figure 3: A schematic of an artificial neuron.

Artificial Neural Networks

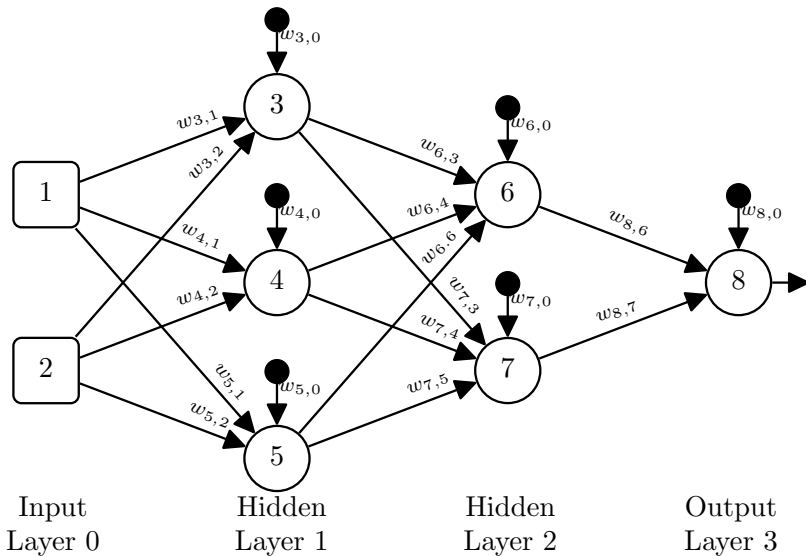


Figure 4: A schematic of a feedforward artificial neural network.

$$\mathbf{z}^{(2)} = \mathbf{W}^{(2)} \mathbf{a}^{(1)} \quad (6)$$

Neural Networks as Matrix Operations

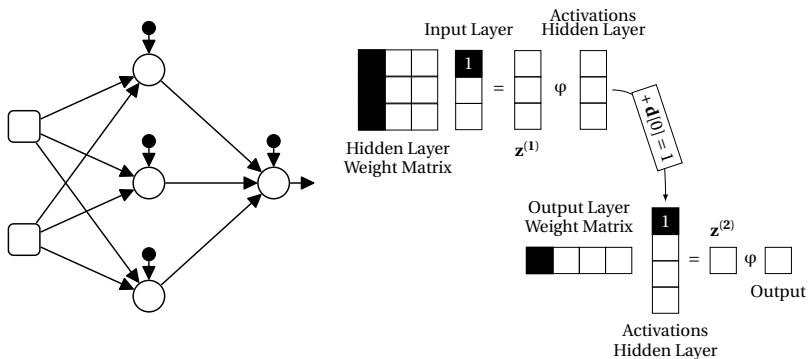


Figure 5: An illustration of the correspondence between graphical and matrix representations of a neural network. This figure is inspired by Figure 3.9 of (Kelleher, 2019).

Neural Networks as Matrix Operations

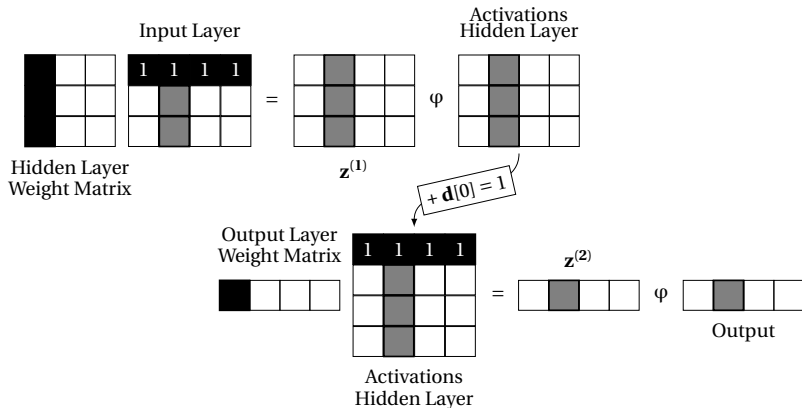


Figure 6: An illustration of how a batch of examples can be processed in parallel using matrix operations.

Why Are Non-Linear Activation Functions Necessary?

$$\mathbf{A}^{(1)} = \mathbf{W}^{(1)} \mathbf{A}^{(0)} \quad (7)$$

$$\mathbf{A}^{(2)} = \mathbf{W}^{(2)} \mathbf{A}^{(1)} \quad (8)$$

$$\mathbf{A}^{(2)} = \mathbf{W}^{(2)} \left(\mathbf{W}^{(1)} \mathbf{A}^{(0)} \right) \quad (9)$$

$$\mathbf{A}^{(2)} = \left(\mathbf{W}^{(2)} \mathbf{W}^{(1)} \right) \mathbf{A}^{(0)} \quad (10)$$

$$\mathbf{A}^{(2)} = \mathbf{W}' \mathbf{A}^{(0)} \quad (11)$$

Why Is Network Depth Important?

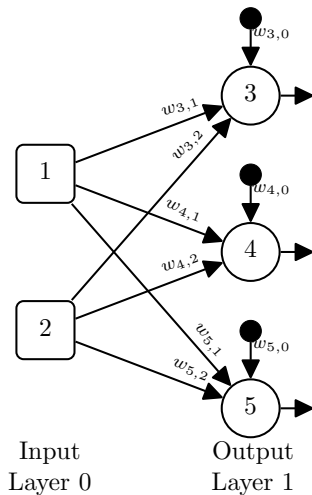


Figure 7: A single-layer network.

Why Is Network Depth Important?

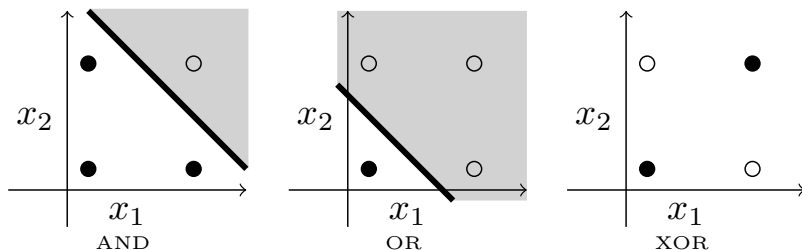


Figure 8: The logical AND and OR functions are linearly separable, but the XOR is not. This figure is Figure 4.2 of (Kelleher, 2019) and is used here with permission.

Why Is Network Depth Important?

$$\mathbb{M}_{\mathbf{w}}(\mathbf{d}) = \begin{cases} 1 & \text{if } z \geq 1 \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

Why Is Network Depth Important?

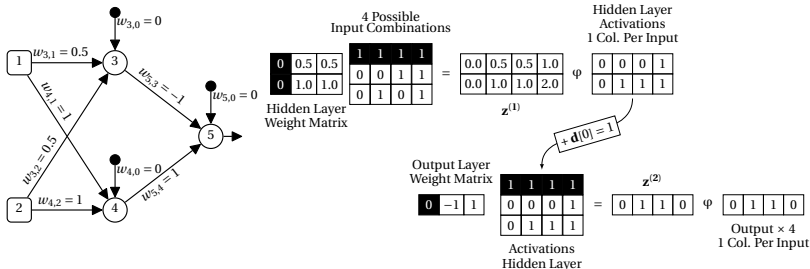


Figure 9: (left) The XOR function implemented as a two-layer neural network. (right) The network processing the four possible input combinations, one combination plus bias input per column: $[bias, FALSE, FALSE] \rightarrow [1, 0, 0]$; $[bias, FALSE, TRUE] \rightarrow [1, 0, 1]$; $[bias, TRUE, FALSE] \rightarrow [1, 1, 0]$; $[bias, TRUE, TRUE] \rightarrow [1, 1, 1]$.

Why Is Network Depth Important?

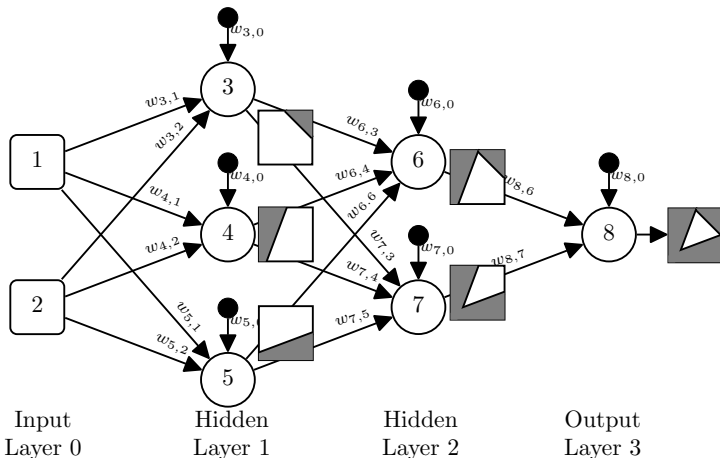


Figure 10: An illustration of how the representational capacity of a network increases as more layers are added to the network. This figure was inspired by Figure 4.2 in (Reed and Marks, 1999) and Figure 3.10 in (Marsland, 2011).

Why Is Network Depth Important?

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Slide Acknowledgment

The slides used in this course are based on the official textbook materials, with modifications made where necessary to suit the course requirements and enhance the learning experience.

Why Is Network Depth Important?

Kelleher, John D. 2019. *Deep learning. Essential knowledge series*. MIT Press.

Marsland, Stephen. 2011. *Machine learning: an algorithmic perspective*. CRC Press.

Reed, Russell D., and Robert J. Marks. 1999. *Neural smithing: supervised learning in feedforward artificial networks*. MIT Press.